MICHAEL MILLAN

Arcata, CA | (213) 326-8756 | millanm488@gmail.com | Portfolio

PROFESSIONAL SUMMARY

Mechanical Engineer (BS, UC Merced) with 2+ years of SolidWorks experience in sheet metal design, structural weldments, and kinematic simulation. Passionate about automotive systems — suspension geometry, chassis design, and mechanical optimization — with hands-on fabrication skills to match. Seeking a design engineering role where technical depth drives real outcomes.

WORK EXPERIENCE

CAD Designer / Design Engineer | O&M Industries | Arcata, CA | *Feb 2025 – Present*

- Translate customer drawings and in-person briefs into production-ready SolidWorks sheet metal and weldment models, generating complete shop drawing packages including flat patterns, BOMs, and fabrication notes.
- Use ProNest for flat pattern nesting and material optimization; collaborate directly with clients on custom one-off projects from initial concept through delivery.
- Introduced 3D scan-to-CAD workflow using Shining3D EinScan, enabling reverse engineering of existing assemblies and improving fit accuracy on custom components.

ENGINEERING PROJECTS

Long Travel Front Suspension System — *2021 F-150 — Personal Engineering Project* | *2024 – Present*

- Built a full kinematic model in SolidWorks from 3D scan data, replicating OEM suspension geometry (caster, camber, toe); ran parametric Design Studies cycling ±9" of wheel travel across multiple steering inputs to optimize outer CV angles and minimize drivetrain binding.
- Tracked CV angle, camber gain, toe change, and bumpsteer across full travel range in Excel — using data-driven analysis to finalize redesigned upper and lower control arm geometry.
- Currently transitioning from validated kinematic model to full 3D component design for fabrication.

2021 F-150 Multi-System Build — *Personal Engineering & Fabrication* | *2023 – Present*

- Designed and fabricated a 3-piece aluminum underbody skid plate system, roll-up bed cover, Molle panel bed storage system with modular tool attachments, and a winch fairlead plate — all developed from Shining3D EinScan scan data through SolidWorks sheet metal modeling to self-performed TIG welding and forming.
- Established a repeatable scan-to-fabrication workflow resulting in precise OEM-fit parts on first build across multiple projects.

EDUCATION

BS Mechanical Engineering | University of California, Merced | | *May 2024*

Welding Technology Certificate | Merced Community College | | *Dec 2024*

TECHNICAL SKILLS

CAD / Design: SolidWorks (CSWA), AutoCAD, Fusion 360 — sheet metal, weldments, assemblies, Design Study, simulation, kinematic modeling

Fabrication & Tools: Shining3D EinScan 3D scanning, ProNest, TIG/MIG/SMAW welding, CNC, metal forming, plasma cutting, 3D printing

Analysis: SolidWorks Design Study, suspension geometry & kinematics, Excel data analysis and charting

Soft Skills: Client-facing design communication, project ownership, cross-functional collaboration, technical documentation